

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is a DoS attack. The logs show a large number of SYN packet requests coming from an unfamiliar IP address, because of which the web server stops responding. This indicates a SYN flood attack which is a type of a DoS attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The three-way handshake consists of these steps:

1. A SYN packet going from the visitor to the web server, requesting for access.
2. A SYN-ACK packet going from the web server to the visitor accepting the connection request.
3. An ACK packet going from the visitor to the web server acknowledging the permission to connect.

When a SYN packet is received by a web server, it reserves some resources for the source to connect. In case of a SYN flood attack, a malicious actor sends a large number of SYN packets to the web server all at once, which causes the server to use all its available resources and hence no resources are left for any other TCP connection requests.

The logs indicate that the web server has been overwhelmed by the large amount of SYN requests and is unable to process or respond to any more requests. This is why a connection timeout error is shown to any new user who attempts to visit the website.